

LESSON PLAN

Topic	Graphs of Motion (KSSM Mathematics Form 4, Chapter 7)
Duration	60 minutes
Learning Mode	Online
Learning Theory	Cognitivism (Sweller's Cognitive Load Theory) & Constructivism
Instructional Model	ADDIE Model

Learning Outcomes

1. Students can identify and differentiate between Distance-Time graph and Speed-Time graph based on slope and y-axis characteristics.
2. Students can interpret motion graphs and explain the movement of an object based on the given graph.
3. Students can apply motion graphs to real-life situations and solve related problems.

Lesson Activities

Description	Activities	Tools	Deliverable
Introduction	Students access Google Sites and view the Session Slide Deck for a visual overview of key concepts. They then observe a dynamic GeoGebra motion simulation to see how slope changes correspond to speed changes in real time.	Canva Slides + GeoGebra	Session Slide Deck
Lecture	Students watch a teacher-made animated video lecture explaining Distance-Time and Speed-Time graphs with visual storytelling using the KTM Komuter journey as a real-life context. Embedded checkpoint questions on Edpuzzle must be answered correctly before proceeding.	Canva + Edpuzzle	Sample Video Lecture File
Activity	Students explore a gamified racing activity on Blooket follows, where correct answers allow students to advance their avatar and reinforcing concepts through play.	Blooket	Mini-Lesson Activity Template
Assessment	Students complete a competitive quiz on Kahoot! identifying graph types	Kahoot! + Quizizz	Self-Assessment Quiz Link

	from images. This is followed by a self-paced Quizizz quiz with mixed question types. Teacher reviews the accuracy report to identify individual learning gaps.		
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Mapping to 5 Domains of Instructional Technology

Domain	Evidence in This Lesson
Design	Learning outcomes, instructional sequence and tool selection aligned to ADDIE model
Development	Canva Slide Deck, teacher-made Video Lecture (Canva), Blooket Mini-Lesson Activity, Quizizz Self-Assessment Quiz
Utilization	Students use GeoGebra, Edpuzzle, Blooket, Kahoot! and Quizizz in a structured learning sequence on Google Sites
Management	All materials compiled and uploaded immediately following the session where teacher monitors engagement and progress via Quizizz accuracy report
Evaluation	Kahoot! competitive quiz + Quizizz self-paced quiz. The accuracy report used for formative assessment and identifying individual learning gaps

AI Declaration

This lesson plan was developed with the assistance of Generative AI tools: Claude (Anthropic) and Gemini (Google). AI-generated responses were compared, critically evaluated and adapted by the student in accordance with USM AI Policy 2026.

Component	Source
<i>Topic selection</i>	<i>100% by student</i>
<i>Tools selection & justification</i>	<i>AI assisted & adapted (60% student, 40% AI)</i>
<i>Learning outcomes</i>	<i>100% by student</i>
<i>Activities design</i>	<i>AI assisted & adapted (60% student, 40% AI)</i>
<i>5 Domains mapping</i>	<i>AI assisted & adapted (70% student, 30% AI)</i>
<i>Table structure & format</i>	<i>100% by student</i>